METRIC
DOD-W-63474A
15 May 1984
SUPERSEDING
DOD-W-63474
31 October 1980

MILITARY SPECIFICATION

WASHERS, SPRING TENSION, METRIC
GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers metrically dimensioned steel, spring tension washers: cylindrically curved, waved and Belleville spring.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

MILITARY

MIL-I-6868 - Inspection Process, Magnetic Particle
MIL-C-81562 - Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited)
DOD-W-63474/1 - Washers, Spring Tension, Cylindrically Curved and Waved, Metric
DOD-W-63474/2 -Washer, Spring Tension, Belleville Spring, Metric

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, Attn: DRSMC-TST-S (D), Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5310

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1312 - Fasteners, Test Methods

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture (Surface Roughness, Waviness and Lay)

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-A684/A684M - Steel, High-Carbon, Strip, Cold-Rolled Soft, Untempered Quality, Standard Specification For

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references, other than specification sheets, cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

3.2 Material. Recycled and reclaimed materials shall be used to the maximum extent practicable. Washers shall be made of spring steel of suitable quality to meet the requirements specified herein.

3.2.1 Spring steel. Spring steel shall conform to any of the steel Grades 1050 thru 1095 (UNS G10500 thru UNS G10950), as specified in ASTM A684/A684M.

3.3 Surface hardness and finish.

3.3.1 Surface hardness. Washers shall have a surface hardness of 43-50HRC (82-85.5HR15N, for washer thickness less than 1 mm).

3.3.2 Protective finish. When specified on the applicable specification sheet, washers shall be cadmium plated either electrodeposited or mechanically deposited.

3.3.2.1 Electrodeposited. Electrodeposited cadmium plating shall be in accordance with QQ-P-416, Type II, Class 2 (7.6 µm thick).

3.3.2.2 Mechanically deposited. Mechanically deposited cadmium plating shall be in accordance with MIL-C-81562, Type II, Class 2 (7.6 µm thick).

3.4 Dimensions. Dimensions and tolerances shall be in accordance with the applicable specification sheet and shall apply after protective finishes.

3.4.1 Edges and burrs. All sharp edges and burrs shall be removed.

3.4.2 Concentricity. The outside diameter (OD) of the washers shall be concentric with the inside diameter within the following limits:

OD (mm)		CONCENTRICITY (µm) 1/
Over	To	
1	3	125
3	6	150
6	10	180
10	18	215
18	30	260
30	50	310
50	80	370
80	120	435

1/ Concentricity dimensions apply to flat spring washers before being formed.

3.5 Surface roughness. Washers shall have a maximum surface roughness of 3.2 um, determined in accordance with ANSI B46.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

3.6 Decarburization. Washers shall be free from decarburization.

3.7 Cracks. Washers shall be free from cracks, flaws and pits in any location. A crack is a clean crystalline fracture passing through or across the grain boundaries without inclusion of foreign elements.

3.8 Performance.

3.8.1 Loading. (Cylindrically curved and waved washers). The washers shall withstand the compression load specified in Table IV and regain its free height within the limits stated on the applicable specification sheet.

3.8.2 Prolonged loading. Cylindrically curved and waved washers shall withstand the compression load specified (see Table IV) for a minimum of 48 hours and regain its free height within the limits stated on the applicable specification sheet.

3.8.3 Load-deflection. Belleville spring washers shall meet the compression load and deflection requirements set forth on the applicable specification sheet.

3.8.4 Permanent set. After one deflection to flat, the Belleville spring washers shall return to their original height within the tolerances of the original height.

3.8.5 Repeated loading. The Belleville spring washers shall withstand the repeated loading test specified in 4.4.3.9 without indication of permanent set in excess of 10 percent of the original free height and without cracking or breaking.

3.9 Workmanship. Washers shall be free of surface contamination, tool marks and other imperfections which may adversely affect useability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Group A and B inspections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

4.2.1.1 Inspection lot. An inspection lot shall consist of all washers covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at any one time.

4.2.1.2 Group A inspection. Group A inspection shall consist of the inspection specified in Table I. The Acceptable Quality Level (AQL) shall apply to each individual defect, not to a group of defects.

4.2.1.2.1 Sampling plan. Statistical sampling and inspection for Subgroup 1 shall be in accordance with MIL-STD-105, Inspection level S-3. Major and minor defects shall be as defined in Table II. Sampling and inspection for Subgroup 2 shall be in accordance with MIL-STD-105, Inspection level S-1.

4.2.1.2.1.1 Rejected lots (Subgroup 1). Rejected inspection lots may be resubmitted for Government acceptance only if the manufacturer performs 100 percent inspection on the washers of the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units and resubmits the lot for quality conformance inspection. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots. Resubmitted lots shall be inspected using the tightened inspection procedure of MIL-STD-105 and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed.

4.2.1.2.1.2 Rejected lots (Subgroup 2). If an inspection lot is rejected, the manufacturer may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection procedure of MIL-STD-105 and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

TABLE I. GROUP A INSPECTION.

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH	SAMPLING PROCEDURE	
Subgroup 1			AQL (Percent defective)	
			Major	Minor
Dimensions	3.4	4.4.1	2.5	4.0
Protective finish	3.3.2	4.4.3.2	—	4.0
Surface roughness	3.5	4.4.1	—	4.0
Subgroup 2			AQL (Percent defective)	
Cracks	3.7	4.4.3.3	1.5	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

4.2.1.2.2 Disposition of sample units. Sample units which have passed all the Group A inspections may be delivered on the contract or purchase order.

TABLE II. CLASSIFICATION OF DEFECTS.

CATEGORY	DEFECT	INSPECTION METHOD
Critical	None defined	
Major	AQL = 2.5 percent defective	
101	Inside diameter, not as specified (see 3.4)	SIE 1/
102	Outside diameter, not as specified (see 3.4)	SIE 1/
103	Evidence of cracks or pits (see 3.7)	Visual
104	Thickness, not as specified (see 3.4)	SIE 1/
105	Free height, not as specified (see 3.4)	SIE 1/
Minor	AQL = 4.0 percent defective	
201	Protective finish, missing or incomplete (see 3.3.2)	Visual
202	Other dimensions, not as specified (see 3.4)	SIE 1/
203	Concentricity, not as specified (see 3.4.2)	SIE 1/
204	Surface roughness, not as specified (see 3.5)	SIE 1/
205	Workmanship, (see 3.9)	Visual

1/ Standard inspection equipment

4.2.1.3 Group B inspection. Group B inspection shall consist of the tests specified in Table III in the order shown. They shall be performed on sample units that have been subjected to and have passed the Group A inspection.

TABLE III. GROUP B INSPECTION.

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH	AQL (PERCENT DEFECTIVE)
Surface hardness	3.3.1	4.4.3.1	1.5
Decarburization	3.6	4.4.3.4	1.5
Load deflection	3.8.3	4.4.3.7	1.5
Loading	3.8.1	4.4.3.5	1.5
Permanent set	3.8.4	4.4.3.8	1.5
Prolonged loading	3.8.2	4.4.3.6	1.5
Repeated loading	3.8.5	4.4.3.9	1.5

4.2.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105. Unless otherwise specified herein, S-1 inspection shall be used.

4.2.1.3.2 Defectives. If the number of defects exceed the number allowed in Table II, the sample shall be considered to have failed.

4.2.1.3.3 Disposition of samples. Sample units which have been subjected to Group B inspection shall not be delivered on the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

4.2.1.3.4 Noncompliance. If a sample fails to pass Group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken Group B inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group A inspection may be reinstituted; however, final acceptance and shipment shall be witheld until the Group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.3 Inspection of packaging. The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with the requirements of PPP-H-1581.

4.4 Methods of inspection.

4.4.1 Visual and dimensional. The washers shall be examined to verify that physical dimensions, surface roughness and workmanship are in accordance with the applicable requirements of 3.4, 3.5, 3.9 and 4.2.1.2.

4.4.2 Material inspection. Material inspection shall consist of certification supporting verifying data that the materials used in fabricating the washers are in accordance with the applicable requirements of 3.2.

4.4.3 Surface hardness and finish inspection.

4.4.3.1 Surface hardness inspection. Samples taken as specified in 4.2.1.3 shall be tested for surface hardness in accordance with MIL-STD-1312, Test method 6 and 3.3.1.

4.4.3.2 Protective finish inspection. Samples taken as specified in 4.2.1.2 shall be inspected for adequacy of plating in accordance with applicable specification of 3.3.2.

4.4.3.3 Magnetic particle (cracks) inspection. Samples taken as specified in 4.2.1.2. shall be subjected to magnetic particle inspection in accordance with MIL-I-6868. There shall be no evidence of cracks or pits as specified in 3.7.

4.4.3.4 Decarburization. Samples taken as specified in 4.2.1.3 shall be microscopically examined at a magnification of 100 diameters. There shall be no evidence of decarburization. The etchant shall be 5 percent nital.

4.4.3.5 Loading. Samples taken as specified in 4.2.1.3 shall be compressively loaded in accordance with Table IV for two minutes. On removal of the load the height of the washers shall be not less than the minimum free height specified on the specification sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

TABLE IV. TEST LOADS FOR CYLINDRICALLY CURVED AND WAVED WASHERS. 2/

WASHER 1/ SIZE DESIGNATION mm	1	1.2	1.4	1.6 1.7 1.8	2	2.3	2.5 2.6	3	3.5	4	5
TEST LOAD (N) MIN	150	200	300	400	650	800	900	1,700	2,200	2,900	4,700

WASHER 1/ SIZE DESIGNATION mm	6	7	8	10	12	14	16
TEST LOAD (N) MIN	6,700	9,600	12,200	19,300	28,000	38,200	52,200

WASHER 1/ SIZE DESIGNATION mm	18	20	22	24	27	30
TEST LOAD (N) MIN	64,000	81,000	101,000	117,000	153,000	187,000

WASHER 1/ SIZE DESIGNATION mm	33	36	39	42 To 52
TEST LOAD (N) MIN	251,000	272,000	325,000	350,000

1/ Corresponds to the nominal bolt/screw diameter sizes intended for use with washer.

2/ These loads do not apply to Belleville washers (DOD-W-63474/2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

4.4.3.6 Prolonged loading. The prolonged loading test applies only to washers with electrodeposited protective finish. Washers, taken as specified in 4.2.1.3, threaded on to a pin and separated from each other by plain washers, shall be compressively loaded in accordance with Table IV for 48 hours without fracturing. On removal of the load the height of each washer shall be not less than the minimum free height specified on the applicable specification sheet.

4.4.3.7 Load deflection. Belleville spring washer samples, taken as specified in 4.2.1.3, shall be deflected to 75 percent of original free height. The resulting load shall be measured to verify conformance with the load and tolerance set forth on the applicable specification sheet.

4.4.3.8 Permanent set. Belleville spring washer samples taken as specified in 4.2.1.3, shall be deflected one time to flat to determine compliance with 3.8.4.

4.4.3.9 Repeated loading. Belleville spring washer samples taken as specified in 4.2.1.3 shall be deflected rapidly 10 times from free height to within .077 mm to .13 mm of solid height to determine conformance with 3.8.5. The load-deflection shall then be performed to determine conformance with 3.8.3.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use. Spring tension washers are intended for use as compression springs where high loads in limited spaces are required. These washers can be used to maintain load or tension in bolted assemblies and to assure proper positioning and tension of ball bearings.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet
b. Applicable specification sheet numbers (see 3.1)
c. Level (degree) of protection in accordance with PPP-H-1581, ordering data (see 5.1)

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AR
Navy - OS

Review activities:
Army - AV
DLA-IS

User activities:
Army - AT
Navy - MC, SH

Agent:
DLA-IS

Preparing activity:
Army - AR

(Project 5310-1253)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-W-63474	Washers, Spring Tension, Metric General Specification For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.